

GEORGIA INSTITUTE OF TECHNOLOGY
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING

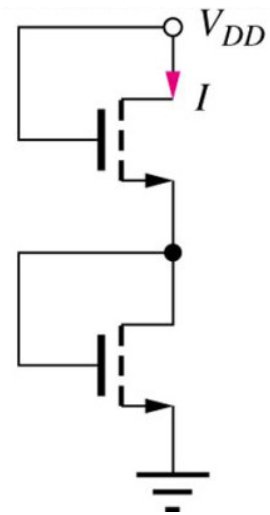
ECE 3040: Microelectronic Circuits

Homework #9

Homework Due Date: Monday, April 27, 2026

1. MOSFET: Circuit Analysis (20 points)

Find the current I in the circuit on the right if $V_{DD} = 10\text{ V}$ and $\lambda = 0$. Both transistors have $W/L = 10/1$, $V_{TN} = 0.75\text{ V}$ and $K'_n = 25\text{ }\mu\text{A/V}^2$. Repeat your calculation for $\lambda = 0.02\text{ V}^{-1}$.



2. MOSFET: PMOS Operating Regions (10 points)

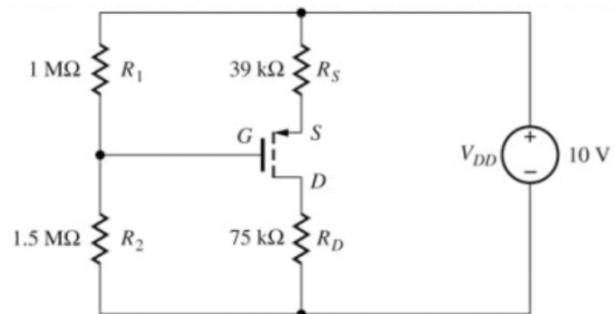
Find the region of operation and the drain current in a PMOS transistor with $W/L = 10/1$, $V_T = -0.75\text{ V}$, $K'_p = 10\text{ }\mu\text{A/V}^2$ for

(a) $V_{GS} = -1.1\text{ V}$ and $V_{DS} = -0.2\text{ V}$

(b) $V_{GS} = -1.3\text{ V}$ and $V_{DS} = -0.2\text{ V}$

2. MOSFET: Q-Point Analysis (30 points)

Find the Q-point for the transistor (a PMOS transistor) on the right. Assume that $V_{TP} = -1\text{ V}$, $W/L = 1/1$ and $K'_p = 25\text{ }\mu\text{A/V}^2$.



4. Common-Source Amplifier (40 points)

Calculate the Q-point and the voltage gain v_o/v_i of the common-source amplifier shown on the right. Assume $R_I = 1\text{ k}\Omega$, $R_1 = 2.2\text{ M}\Omega$, $R_S = 22\text{ k}\Omega$, $R_D = 18\text{ k}\Omega$, $R_3 = 470\text{ k}\Omega$, $K_p = 400\text{ }\mu\text{A/V}^2$, $V_T = -1\text{ V}$, $\lambda = 0.02\text{ V}^{-1}$, $V_{DD} = 18\text{ V}$.

